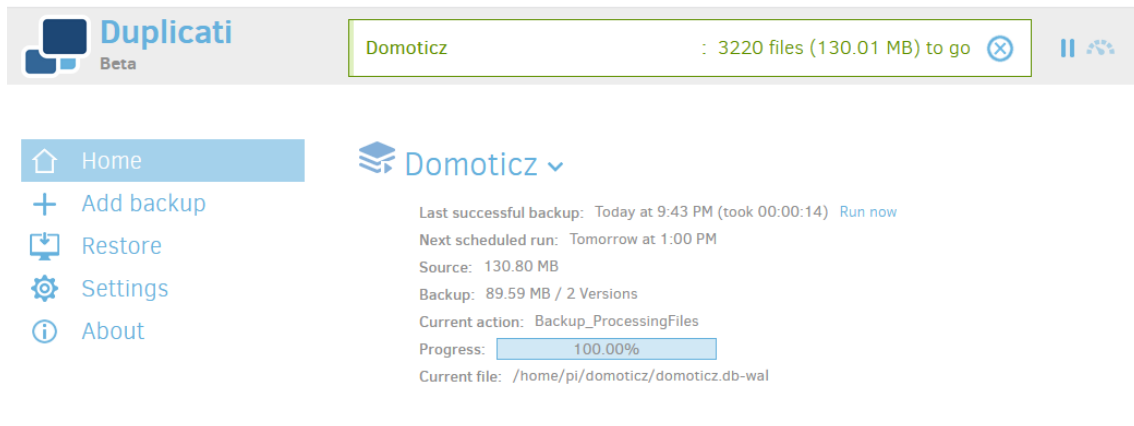




Backup Domoticz with Duplicati

[Domoticz](#) March 29, 2020

Step-by-step how-to guide install and configure **Duplicati** with the latest **Raspbian Buster** image and **Domoticz**.

Duplicati is a fantastic and very intuitive backup solution that support a wide range of backup targets. Combined with a webinterface, this makes it perfect for a backup solution for our beloved Domoticz installation.

Duplicati supports not only various online backup services like OneDrive, Amazon S3, Backblaze, Rackspace Cloud Files, Tahoe LAFS, and Google Drive, but also any servers that support SSH/SFTP, WebDAV, or FTP.

It does however need the mono framework as it is mainly written for the Windows platform and this makes it a bit heavy for Linux. Nonetheless, it's one of my favorite open-source backup products.

Tip: Duplicati is more about making a backup of the most important files. If you are looking for a complete system backup solution, perhaps this is more what you are looking for:

<https://www.hackviking.com/single-board-computers/raspberry-pi/automated-raspberry-pi-backup-complete-image/>

This how-to is part of a bigger series of Domoticz how-to's on sancla.com!

This tutorial has been verified with:
Domoticz 2020.1

Prerequisites

- Running Domoticz installation with stable Raspbian Buster release and SSH access. See my previous post for a how-to:
<https://www.sancla.com/domoticz/raspberry-pi-4-with-domoticz/>

Tested with

- Raspberry Pi 4 (MEM 2GB with 16GB sd-card)
- Raspbian Buster Lite 4.19, Februari 2020
- Domoticz Stable 2020.1 (compile date 22-3-2020)
- Duplicati BETA [v2.0.5.1-2.0.5.1_beta_2020-01-18](#)

Step 1: Install the Mono Framework

This step can take a bit of time...

```
sudo apt install mono-complete ca-certificates-mono -y
```

Step 2: Sync the certificates

```
sudo cert-sync /etc/ssl/certs/ca-certificates.crt
```

Step 3: Install Duplicati

```
wget https://github.com/duplicati/duplicati/releases/download/v2.0.5.1-2.0.5.1_beta_2020-01-18/duplicati_2.0.5.1-1_all.deb  
sudo apt-get install ./duplicati_2.0.5.1-1_all.deb -y
```

Step 4: Install missing dependencies

Sometimes Raspbian is still missing some required dependencies.
Just to be sure, run the following command so we are not missing out on any:

```
sudo apt -f install -y
```

Step 5: Configure the service file

We need to edit the service file in order to make it possible to run this as a service. So go open the file with the nano text editor with the following command:

```
sudo nano /etc/systemd/system/duplicati.service
```

```
GNU nano 3.2 /etc/systemd/system/duplicati.service

[Unit]
Description=Duplicati web-server
After=network.target

[Service]
Nice=19
IOSchedulingClass=idle
EnvironmentFile=-/etc/default/duplicati
ExecStart=/usr/bin/duplicati-server $DAEMON_OPTS
Restart=always

[Install]
WantedBy=multi-user.target
```

Copy-paste the following configuration and save the file.
To exit and save the changes, press CTRL+X first and confirm to save the file with “Y”).

```
[Unit]
Description=Duplicati web-server
After=network.target

[Service]
Nice=19
IOSchedulingClass=idle
EnvironmentFile=-/etc/default/duplicati
ExecStart=/usr/bin/duplicati-server $DAEMON_OPTS
Restart=always

[Install]
WantedBy=multi-user.target
```

Step 6: Configure the initscript file

Next is to edit the initscript file with the following command:

```
sudo nano /etc/default/duplicati
```

```
GNU nano 3.2 /etc/default/duplicati

# Defaults for duplicati initscript
# sourced by /etc/init.d/duplicati
# installed at /etc/default/duplicati by the maintainer scripts

#
# This is a POSIX shell fragment
#

# Additional options that are passed to the Daemon.
DAEMON_OPTS="--webinterface=any --webinterface-port=8200 --portable-mode"
```

It should resemble the below configuration. You have to add the last “DAEMON_OPTS...” line to the file and save the changes. To exit and save the changes, press CTRL+X first and confirm to save the file with “Y”).

```
# Defaults for duplicati initscript
# sourced by /etc/init.d/duplicati
# installed at /etc/default/duplicati by the maintainer scripts

#
# This is a POSIX shell fragment
#

# Additional options that are passed to the Daemon.
DAEMON_OPTS="--webinterface=any --webinterface-port=8200 --portable-
mode"
```

Step 7: Enable Duplicati service

Now its time to enable and start the service.
Execute the following commands:

```
sudo systemctl enable duplicati.service
sudo systemctl daemon-reload
sudo systemctl start duplicati.service
sudo systemctl status duplicati.service
```

You should now have Duplicati up-and-running and see the below results of a running service:

```

pi@raspberrypi:~ $ sudo systemctl enable duplicati.service
Created symlink /etc/systemd/system/multi-user.target.wants/duplicati.service →
/etc/systemd/system/duplicati.service.
pi@raspberrypi:~ $ sudo systemctl daemon-reload
pi@raspberrypi:~ $ sudo systemctl start duplicati.service
pi@raspberrypi:~ $ sudo systemctl status duplicati.service
● duplicati.service - Duplicati web-server
   Loaded: loaded (/etc/systemd/system/duplicati.service; enabled; vendor preset
   Active: active (running) since Fri 2020-01-03 20:50:44 CET; 3s ago
 Main PID: 20374 (mono)
    Tasks: 16 (limit: 4035)
   Memory: 23.4M
    CGroup: /system.slice/duplicati.service
            └─20374 DuplicatiServer /usr/lib/duplicati/Duplicati.Server.exe --web
              └─20381 /usr/bin/mono-sgen /usr/lib/duplicati/Duplicati.Server.exe --
Jan 03 20:50:44 raspberrypi systemd[1]: Started Duplicati web-server.
lines 1-11/11 (END)

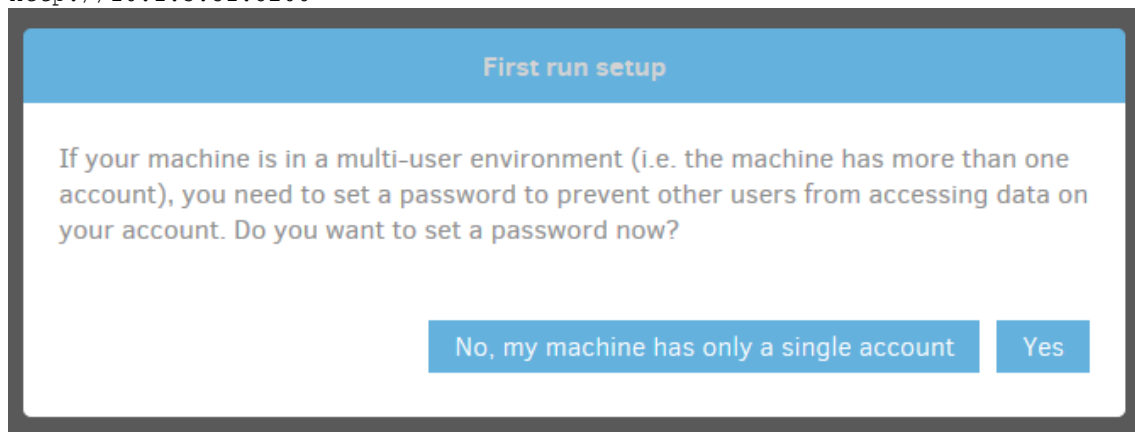
```

Duplicati service is running correctly

Step 8: Open the Duplicati web-gui

Next is to navigate to the web interface/gui. It is configured in the initscript file to run on port 8200. In my case, the IP address of my Raspberry Pi is 10.1.3.51. So in order to reach the web gui, I have to open the following address in my browser:

<http://10.1.3.51:8200>



The first time you open the Duplicati interface, choose the first option “No, my machine has only a single account”.

You should now be able to configure a backup job!

A couple of tips when creating a backup job (see screenshots below for further guidance):

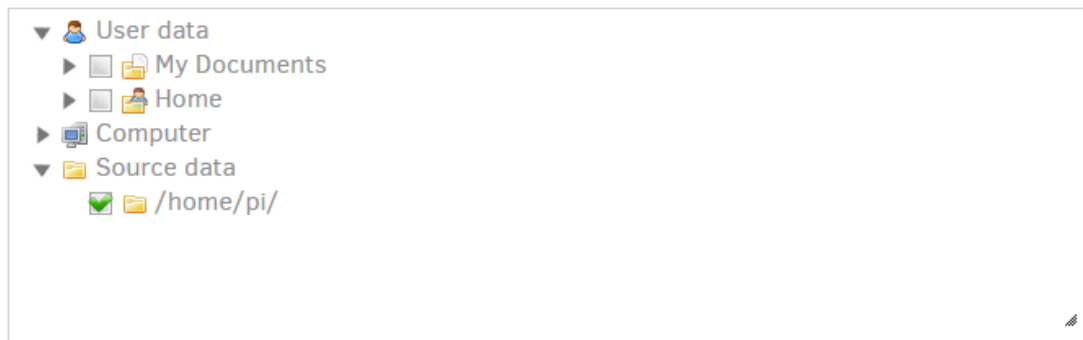
- **Include one of these directories:**
 - **/home/pi**
 - **/home/pi/domoticz**
- **Enable automatic backup in the Domoticz setup menu so you get consistent backups of your database.**
- **Keep your encryption password in a safe place that you can find back should you ever need it (or just do not use an encryption password)!!**
- You can also run scripts in advance or after the backup
- Configure backup retention to automatically clean up old backups.

- Enable e-mail notifications to keep track of your backups.
See this forum post for the proper steps using Google Mail:
<https://forum.duplicati.com/t/how-to-configure-automatic-email-notifications-via-gmail-for-every-backup-job/869>

Source data



☐ Show hidden folders



Add a path directly

Add path

Filters

Exclude

< Previous

Next >

- ***Include the pi home directory (or the lower level domoticz directory)***



General options

Remote volume size

This option does not relate to your maximum backup or file size, nor does it affect deduplication rates. [See this page before you change the remote volume size.](#)

Backup retention	Keep a specific number of backups
	7

Once there are more backups than the specified number, the oldest backups are deleted.

Advanced options ^

Add advanced option

< Previous Save

Enable Backup retention or the pre-backup script option

General options

Remote volume size

Backup retention

Advanced options

Add advanced option

zip-compression-method: Sets the Zip compression method
zip-compression-zip64: Toggles Zip64 support
Run script
run-script-after: Run a script on exit
run-script-before: Run a script on startup
run-script-before-required: Run a required script on startup
run-script-log-filter: Log message filter
run-script-log-level: Defines a log level for messages
run-script-result-output-format: Selects the output format for results
run-script-timeout: Sets the script timeout
HTTP report module
send-http-any-operation: Send messages for all operations
send-http-extra-parameters: Extra parameters to add to the http message
send-http-level: The messages to send
send-http-log-filter: Log message filter

< Previous Save

You can find the script option under Advanced Options

The screenshot shows the Domoticz V4.4.0717 System Setup page. The top navigation bar includes tabs for Dashboard, Switches, Scenes, Temperature, Weather, Utility, and Setup. Below this, a secondary navigation bar contains System, Log History, Notifications, Email, Meters/Counters, Floorplan, Other, and Backup/Restore. The main content area is titled 'System Setup' and contains several configuration sections: User Interface (Language: English, Theme: default), Location (Name: Domoticz, Latitude, Longitude), Dashboard (Mode: Normal, Allow Widget Re-Order), Mobile (Mode: Mobile), Website Protection (Username, Password, Authentication: Login Page), Security Panel (Password, Delay: 30), Light/Switch Protection (Password), Local Networks (no username/password), Remote Shared Port (For Remote Domoticz clients: 6144), Software Updates (Check for Updates), and Automatic Backup (Enable automatic backup). A red arrow points from the 'Dashboard' tab to the 'Automatic Backup' checkbox.

• ***Enable the default Domoticz Automatic Backup solution!!!***

Should you need to restore...

If you ever need to restore, you can do so easily by clicking on the job and selecting to restore files, as seen in the screenshot below.

Follow the steps with these instructions:

<https://duplicati.readthedocs.io/en/latest/03-using-the-graphical-user-interface/#restoring-files-from-a-backup>



Operations:

[Run now](#) [Restore files ...](#)



Configuration:

[Edit ...](#) [Export ...](#) [Delete ...](#)

Advanced:

[Database ...](#) [Verify files](#) [Compact now](#) [Commandline ...](#)

Reporting:

[Show log ...](#) [Create bug report ...](#)

Last successful backup: Today at 9:43 PM (took 00:00:21) [Run now](#)

Next scheduled run: Tomorrow at 1:00 PM

Source: 130.80 MB

Backup: 89.90 MB / 4 Versions

Restore files with Duplicati

Worst-case scenario

However, in the worst-case scenario, a F.U.B.A.R. situation, you don't have a running pi anymore. For example, your SD-card or USB stick died...

First 2 steps are to get your pi running again with a fresh installation of raspbian and Domoticz. Next is to install Duplicati again the guide below. Another option is to just install Duplicati on your own system to retrieve your files. See this link for more information and instructions

<https://duplicati.readthedocs.io/en/latest/03-using-the-graphical-user-interface/#restoring-files-if-your-duplicati-installation-is-lost>

Further documentation

For more information on Duplicati, how to create a proper backup job and other documentation, you can visit this site:

<https://duplicati.readthedocs.io/en/latest/>